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The Three Pillars of Empiricism (Scrum)



Hiren Doshi

(India)



December 4, 2016

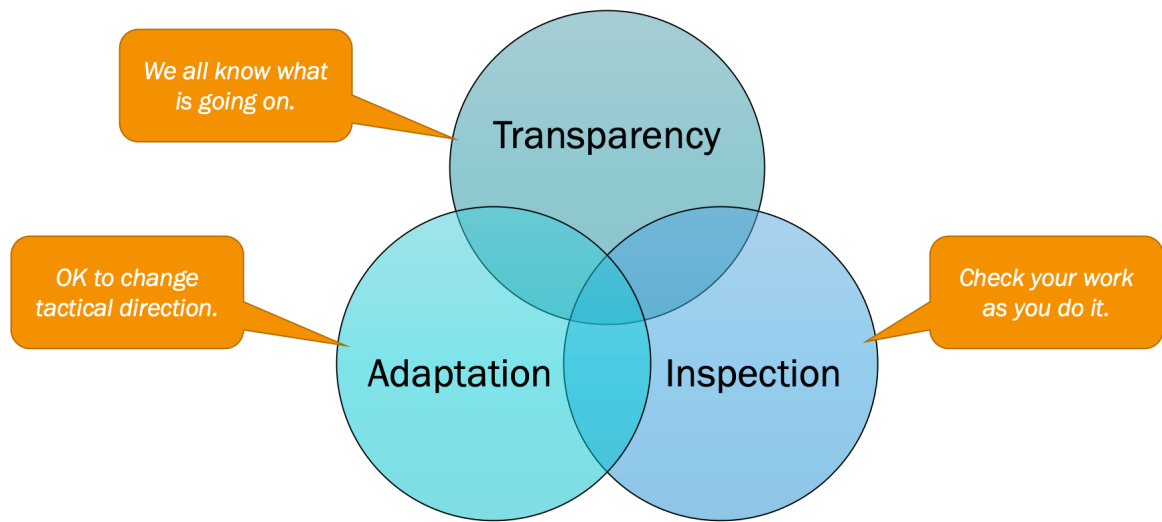
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Empiricism means working in a fact-based, experience-based, and evidence-based manner. Scrum implements an empirical process where progress is based on observations of reality, not fictitious plans. Scrum also places great emphasis on mind-set and cultural shift to achieve business and organizational Agility.

The three pillars of empiricism are as follows:



- **Transparency:** This means presenting the facts as is. All people involved—the customer, the CEO, individual contributors—are transparent in their day-to-day dealings with others. They all trust each other, and they have the courage to keep each other abreast of good news as well as bad news. Everyone strives and collectively collaborates for the common organizational objective, and no one has any hidden agenda.
- **Inspection:** Inspection in this context is not an inspection by an inspector or an auditor but an inspection by every- one on the Scrum Team. The inspection can be done for the product, processes, people aspects, practices, and continuous improvements. For example, the team openly and transparently shows the product at the end of each Sprint to the customer in order to gather valuable feedback. If the customer changes the requirements during inspection, the team does not complain but rather adapts by using this as an opportunity to collaborate with the customer to clarify the requirements and test out the new hypothesis.
- **Adaptation:** Adaptation in this context is about continuous improvement, the ability to adapt based on the results of the inspection. Everyone in the organization must ask this question regularly: Are we better off than yesterday? For profit-based organizations, the value is represented in terms of profit. The adaptation should eventually relay back to one of the reasons for adapting Agile—for example, faster time to market, increased return on investment through value- based delivery, reduced total cost of ownership through enhanced software quality, and improved customer and employee satisfaction.

Scrum works not because it has three roles, five events, and three artifacts but because it adheres to the underlying Agile principles of iterative, value-based incremental delivery by frequently gathering customer feedback and embracing change. This results in faster time to market, better delivery predictability, increased customer responsiveness, ability to change direction by managing changing priorities, enhanced software quality, and improved risk management.

This is one of the topics I covered in my book - "[Scrum Insights For Practitioners: The Scrum Guide Companion](#)". Happy reading!

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Comments (77)

Alan Larimer

11:17 pm December 5, 2016

Paraphrased most valuable sentence that all should remember: "Scrum works not because of its rules, but because, as a predecessor, it adheres to the underlying values and principles of the Manifest for Agile Software Development."

Florian Sauter

08:45 am December 21, 2016

Thank you for this post. Everyone reading this should also check out the of The House of Scrum from Gunther Verheyen - I recommend it because it highlights the fact that transparency is

things.

Jeff	08:40 pm April 30, 2018 Is it me or are these scrum/agile terms simply new phrases/nouns for things we already know in engineering??? Seems more like a cult rather than anything all that useful.
Eric Naiburg	05:54 pm May 21, 2018 Seeing how Scrum is over 22 years old, I wouldn't call them new phrases or nouns. Many have been applied to engineering and vise versa. Having things solidified in a framework, explained how they work together and interpreted is more important than the term alone.
Jeff	02:39 pm May 22, 2018 While they are 22 years old in some circles I never once heard them throughout high school or college. I didn't start to see them demanded from employers until 5-8 years ago. Most people already do those things it's just changing nouns, which most will pick up on within a few weeks...but apparently employers use it as a barrier and reason to deny job hunters who haven't been in college the past 5-10 years.
Cathi Quinn Fitzharris	01:26 am November 2, 2018 i want to take the scrum master certification. what are recommended readings
udai	01:13 pm November 11, 2018 1. Make you current understanding transparent: https://www.scrum.org/index... 2. Inspect what you understand and what you do not understand 3. Adapt. Exploit what you do understand and remove what you do not understand Repeat
Nepolion Nandy	12:08 pm December 20, 2018 Hi Guys, Could anyone please answer this below? "What is the Scrum master's daily routine starts from morning till end of working time, describe the full day with time schedules and in sequence?" Pre requisite - 6 yrs exp scrum mater and partially matured team. Regards, Nepolion Nandy.
Robert-Alexandru Oros	09:28 pm January 3, 2019 Small correction "Scrum works not because it has three roles, five events, and three artifacts but because it ..." It has 4 events, as the guide states that the Backlog refinement is not an event, but more of an ongoing process.
Dicle Akay	08:33 am January 23, 2019 Thank you for the post. I'd like to add that; I think one of the most important result of Agile principles and its 3 pillars is motivated team members. We are periodically measuring team happiness metric and teams are much more happier about their code quality, close relationships with customers, the feedbacks, self-organization and continuous improvement areas. Motivated team members results in better transparency, adaptation and inspection in organizations.
Peter Nieuwveld	10:17 am January 26, 2019 The Sprint itself is also an event, so there are 5 Scrum events - the Sprint, Sprint Planning, Daily Scrum, Sprint Review & Retrospective. And indeed refinement is an ongoing process, not an official event.
Jean-Marc Slatter	03:26 pm February 11, 2019 Did what all others have suggested but also got Scrum Narrative and PSM Exam Guide by Mohammed Musthafa Soukath Ali which really helped. I passed the first attempt but most important is to use what you study and understand concepts not just answers.
NOVAGuy	04:26 am March 16, 2019 So true! Alas the entire Scrum/Agile theory/books expect team members somehow to be motivated by nature, happy, people who feel really bad they couldn't do more work for the company and always willing to work more including weekends. (communist Russia or China???) While Agile mentions 1 day or more for relaxation after the sprint, Scrum goes from iteration to iteration with zero-zero break

companies... this is what I have noticed in all of them:

Once scrum/agile is introduced, all developers want to be managers. As long H1bs continues to flow in, Agile teams have the Green Card as motivation...

Another flaw I see day-to-day activities with scrum, due to not recognizing titles within the team it's f. impossible to ask anyone to check/do anything if it is 1" above their work, e.g "clean pdb files", or "where is your DoD for your task.." or ask a tester "why no fortify for this sprint?"....

Their reply is: "You can figure out yourself, not in my board" If anyone tries to bring it in retrospective or daily scrum meetings for any improvement, fractions go personal while company's management sits back and watch theater how people ruin their lives fueled by humans vices... (oh agile thrives on it)

If cheap labor (h1b) is limited, corporations will increase salaries and increase the productivity in anyway they can (agile/scrum/xp).... and \$\$\$ is the main motivation.

PSM1, PSPO1

NOVAGuy

📅 04:45 am March 16, 2019

Also from definition, scrum does not have a fixed number of artifacts.

By definition: "Scrum’s artifacts represent work or value to provide transparency and opportunities for inspection and adaptation."

- 1. Product backlog
- 2. Sprint backlog
- 3. Increment

are the main artifacts.

Rightfully, MP recognizes 6, adding three more:

- 4. Definition of “Done”
- 5. Monitoring Progress towards a Goal
- 6. Monitoring Sprint Progress

Matt Kipper

📅 01:50 pm May 21, 2019

Scrum and Agile provide management with a fresh-sort-of approach to product development and potentially-gain better validated solutions thanks to the customer checks, but there are plenty of flaws which we need to accommodate in order to make a sensible end-to-end development:-

- 1: No tester role. Really?! So you want to make the developers, the creators, into the destroyers... yeah, sure. Dream on.
- 2: Cross functional roles included in the Development team? So, we have everybody we need in Development to avoid help from 'outside'? So, a body from Procurement, one from legal, one from HR, one from Audit etc..... and we need to limit to 9 people?
- 3: Respect, Openness and Trust? Have these people worked in the real world? This is like the 10 commandments.... Sorry, everybody sins, which starts with politics and ends with defensive emails sapping the who ethos....
- 4: Definition of Done - created by the Development team? Really? That with no Quality Manager role in scrum is a recipe for mess. Developers don't even know what the company needs them to provide. A Quality Manager would be able to guide them to Development Organisation standards including evidence of 'done' with a rational and evidence of what was done and why he/she thinks it's good.
- 5: Timebox/Sprint length always 2 weeks regardless of the needs of the environment e.g. SAFe requiring multiple scrum team output integration test time including bug fixing....

The solutions require experienced pragmatic people backed up with a Quality Manager to prepare them for the dreaded external/internal audits.

Jeff is correct with his observation that these new approaches/frameworks and standards for this, that and others are just being used by hirers to select the younger-but-experienced over older-but-super-experience BSc Computing Science bods.

But whatever new approaches to development we have and get, the bottom line is the right people need to be in power. An engineering person in command, and things tend to go well. Put sales/marketing in command, hmm... time to leave...

Greg Gidney

📅 05:03 am July 22, 2019

@mattkipper:disqus , It's called "integrity". Perhaps the teams you worked with had poor ethics, that doesn't mean all teams or people do.

Tom

📅 01:31 am August 16, 2019

awesome share

Amogh Joshi

📅 10:50 am September 30, 2019

What is the difference between 5. and 6. Is it like release burndow and sprint burndown. You might need the 7. Definition of "Ready" too which should not be mandatory, but it should be recognized that without it, we may have to struggle to continuously improve.

NOVAGuy

📅 08:19 pm September 30, 2019

I see what are you saying.... By definition, artifact represents work or value, regardless related to a sprint goal or not. Sprint includes more than a goal as building a SD/CT model which is

"Monitoring" in itself the way I see it, it should NOT be an artifact. The progress towards the sprint (or goal), only if its outcome provides a value (as a failure or as a success), then it is an artifact and the product owner should accept it.

NOVAGuy

📅 10:12 pm September 30, 2019

You are right, it's a cult! I have been doing agile for last 15 years. No two scrum masters have the same understanding... and all go with the same cult excuse: "The others are not doing it right, but I will do the right scrum..."

Management wants it, it gives them a sense of security and control, yet they can't explain why they can't get more sales....

I got the certifications, 4 of them smoothly from scrum.org.

The only good thing about scrum certifications is that now scrum master can't B.S. too much, at least not in my face.... however, no one is willing to sacrifice the paycheck or the career fighting for the "honor of agile" so the B.S. is floating in our faces, daily, masquerading as SAFe

PSP I, PSD I, PSPO I, PSP I

NOVAGuy

📅 10:40 pm September 30, 2019

"Timebox/Sprint length always 2 weeks regardless of the needs of the environment e.g. SAFe "

Yeah, they are becoming so ridiculous, Sprint goals are becoming adding a button and an event handler with exceptions....

NOVAGuy

📅 12:59 am October 1, 2019

All do know that it not the scrum which is working, it is the debt that force us to work - that is indeed what really is working. As such, we swallow everything given to us deep in our throat and there is nothing that we can do about it.

At that moment the project manager gets in the scrum meetings , monitoring charts, that turns all into a survival political game!

Everyone becomes pragmatic programmer, cut where can be possible cut, finish the deadline, get paycheck today, no one knows what tomorrow brings.

Sounds like cheap Chinese toys? That's the philosophy which those toys are build too!

NOVAGuy

📅 01:02 am October 1, 2019

Make sure you don't think out of the box
1 -2 -3; 1 -2 -3; 1 -2 -3... Agile!

Alan Larimer

📅 06:45 pm October 1, 2019

Welcome to the agile software development philosophy. Please take time to research further as you have highlighted many of the common misunderstandings that newcomers have to the paradigm. It also appears that you have mistaken several common practices or loosely Scrum-based approaches such as S_Fe. (SAFe is traditional and not agile in nature.)

1. Developer is a term used for each individual on the Development Team; it is not a job title from the HR group. There may be technical writers, artists, quality assurance specialists, customers, etc.

2. See above, however, only the roles that are necessary to produce the product Increment are a part of the Development Team. HR, marketing, sales, accounting, etc are a part of the supporting organization.

3. Many organizations, not limited to professional ones i.e. families and friend groups, are full of those who do not uphold good values. Adhering to the values in The Scrum Guide is a must. Organizations and individuals who cannot need to be addressed or move away from the paradigm.

4. See the first. Teams and products are stronger when sharing the expectation for quality. For example, software engineers are encouraged to use Test Driven Design/Development (TDD), collaborate on automated testing approaches and maintenance, etc.

5. This is definitely confusing some specific methodoly directives with the framework of Scrum. Scrum requires no more than one-month Sprints. One week, eight days, three weeks, twenty nine days, etc. are all acceptable within the Scrum framework.

If you believe in the traditional, top-down, boss and workers, command-and-control paradigm then the Scrum framework and the agile product development philosophy may not be for you. It is a significant shift in mentality.

Alan Larimer

📅 06:49 pm October 1, 2019

The Scrum Guide. That's all that should be needed. Supplemental materials must come from the authors of the definition. Cohn, Cuttler, and many others promote what is essentially rebranding of traditional approaches or falsely promote specific practices as requirements of the Scrum framework. <https://scrumguides.org/>

Alan Larimer

📅 06:56 pm October 1, 2019

<https://scrumguides.org/>) and, as with the rest of the framework, is not schedule driven. Cognitive and creative processes are often approached using schedule-driven paradigms primarily developed in manufacturing. There was a time when risk-based was the new paradigm and it is still useful in some areas. The creation of the Manifesto for Agile Software Development (<http://agilemanifesto.org/>) was built upon the experimental processes of several who were intent on improving the multitude of failures of product development due to the traditional paradigms.

Alan Larimer	06:57 pm October 1, 2019 Not at all, but that is how they are commonly applied. http://agilemanifesto.org/
Froggy	07:39 am November 11, 2019 One of the key element to success for Agile projects is communications, especially when you have a multi-cultural multi-skilled team. It applies to all levels of the project, and is fundamental to achieve success.
Manuel PAYET	03:53 pm November 18, 2019 It seams that : https://s3.amazonaws.com/sc... is broken... This XML file does not appear to have any style information associated with it. The document tree is shown below. <error> AccessDenied <message>Access Denied</message> <requestid>BE57FD6FAC168969</requestid> <hostid> AN1FIHkujGRANceIPnkgmBg2MR5CREN8gcJqEEW1IpCh+Dgfm73CQbFTOXbSS3k91INJSc3b5zM= </hostid> </error>
SS	02:14 am February 26, 2020 Hello, I want to take up certification PSP I and II, could you share what all i need study/prepare please Appreciate your guidance thanks
NOVAGuy	06:25 pm March 24, 2020 Sorry for the late reply, I just noticed it that. There is no secret really, I just took the certifications in this logical order PSM1, PSD, PSPO I and the last one from Scrum was PSP. (now I added SAFe) Unfortunately there is no many PSP preparation tests (if any), but recently some may be available at Udemy.com. All I had to do is to read the materials listed on Scrum.org suggerstions, they are there for a reason. If you are going for PSP II, I am not in that position to help, I'm not that good, but If I had to go for it, I would read whatever scrum.org suggests and be active on-line forums for a month or so. Good luck.
Alex E	11:57 am May 2, 2020 Hello, You say "Inspection in this context is not an inspection by an inspector or an auditor". But Scrum Guide says "Inspections are most beneficial when diligently performed by skilled inspectors at the point of work". As I'm preparing for PSM1 assessment I would like to know which assertion to follow ? Regards, Alex
Shaukat Shah	05:40 pm May 8, 2020 Concise and great. Let me try to write summary. Empiricism or objective view point stands on the pillars of transparency, inspection and adoption. Agile encourages to share desirable and undesirable performance, this leads to "transparency" resulting in timely "inspection" (evaluation) and change (adoption).
Azeem Sohail	07:21 am October 24, 2020 Did you know about the history of telescope ?
Shelby Zhang	07:05 pm October 24, 2020 I just have to say that I've read so many of your comments on different posts, and you are incredible. I've learned more from reading your in-depth and detailed comments for the past ten minutes than I have reading through these posts. Thank you for your answers and explanations.
Alan Larimer	07:42 pm October 26, 2020 Thank you for the kind words! I am glad that my contributions have been helpful.
Govindraj Tungenwar	03:23 pm November 3, 2020 Nice article



HL	03:53 pm October 13, 2021 Exactly right! That is the key so many shops (and so many consultants) completely overlook.
Sanjay Chadha	07:43 pm November 11, 2021 I recommend, keep taking open tests - there are several. do not rely just on scrum.org do lot more.
Samuel Enoch	08:18 am December 3, 2021 Very simple but clear explanation of how the three pillars of Empiricism operate.I strongly believe Transparency,Inspection and Adaptation are so closely interwoven that once any of the three is removed or undermined,the whole process will collapse like a pack of cards.
Ayan Ghosh	03:07 pm December 4, 2021 Good explanation. Simple but to the point. Thanks for posting it.
abgorjhouli	06:18 am February 27, 2022 안전놀이터 Seo Han-sol started leisurely with six consecutive hits in two innings. It was 66 points, 20 points difference from the remaining three players. However, he later made a lot of hits and eventually survived
abgorjhouli	06:18 am February 27, 2022 먹튀검증 It was 'new intelligence'. Shin Jin-seo, a 9th grader, also beat Ke Jie, a 9th grader. It was a showdown between Korea and China. The scenario of winning a come-from-behind victory in Shin Jin-seo's
abgorjhouli	06:18 am February 27, 2022 먹튀검증커뮤니티 South Korea's final runner Shin Jin-seo 9 dan won three consecutive games against China's final runner Ke Jie 9 dan in the 13th round of the 23rd Nongshim Shin Ramyun Cup World Baduk Championship in
abgorjhouli	06:18 am February 27, 2022 안전놀이터 Shin Jin-seo 9 dan faced his old rival Ke Jie 9 dan online amid high evaluation that it was a de facto championship match. It was the match between No. 1 in Korean rankings for 26 consecutive months
abgorjhouli	06:18 am February 27, 2022 안전놀이터순위 Shin Jin-seo, a 9-dan player who started as a white player by covering the stone, began to upload the winning rate graph near the 40th number. In a neck-and-neck race from the right to the idol, it soared
Junior Santana	09:22 pm March 29, 2022 Todos os eventos do scrum , criam a possibilidade de manter os 3 princípios dentro de um ambiente empírico
Túlio Carvalho	08:28 pm April 23, 2022 Muito bom.
sandeep gupta	05:21 pm May 3, 2022 Do we really inspect at the end of the sprint....? 1 2 > »

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